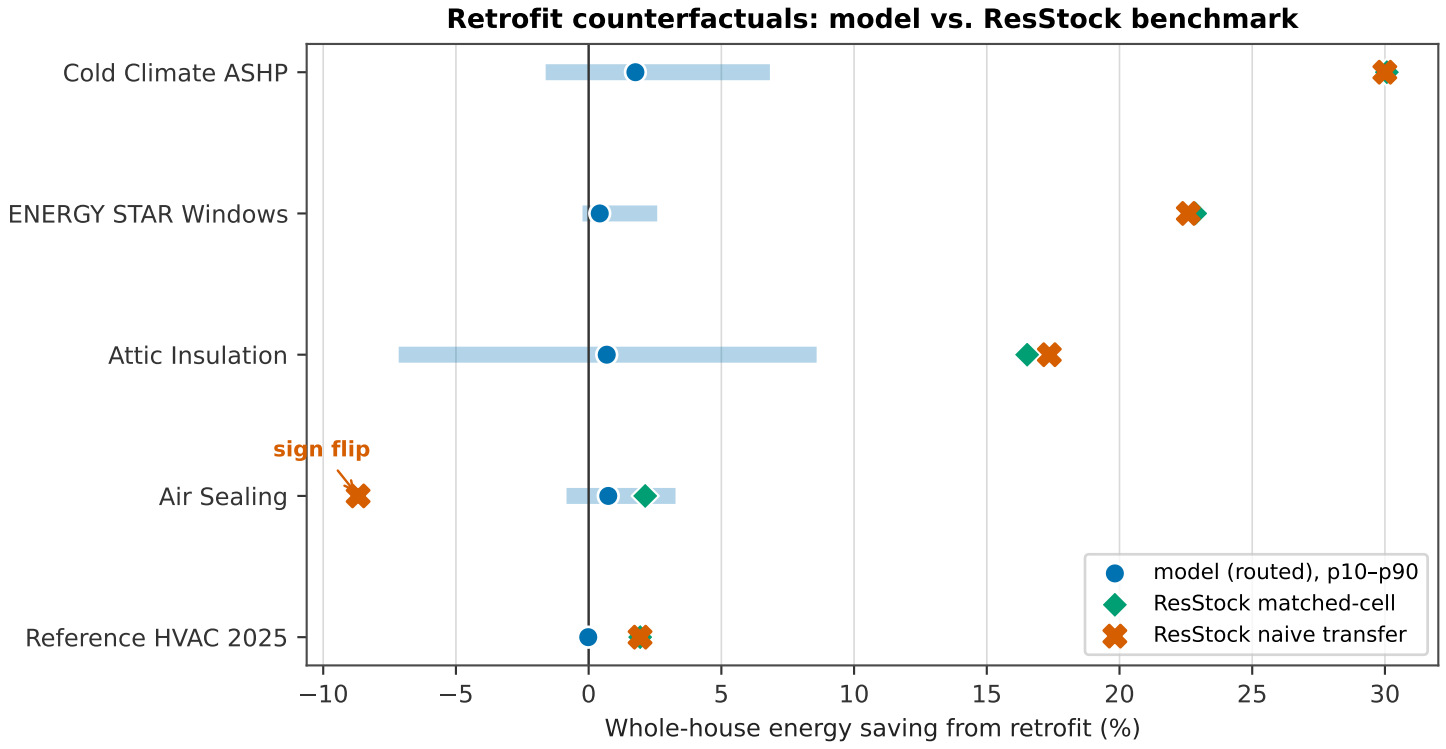
Retrofit counterfactuals: model vs. ResStock benchmark
Cold Climate ASHP
ENERGY STAR Windows
Attic Insulation
sign flip
Air Sealing
Reference HVAC 2025
model (routed), p10–p90
ResStock matched-cell
ResStock naive transfer
−10
−5
0
5
10
15
20
25
30
Whole-house energy saving from retrofit (%)